Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Need for statistics: purposes, and the collection and tabulation of data. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	9
Topic	Need for statistic
Subtopic	Need for statistics: purposes, and the collection and tabulation of data
Theme	—
Curriculum objectives	• explain the purpose of statistic • analyse and interpret statistical data presented in tables
Previous knowledge	• Students are familiar with counting and recording information.
Lesson sequence	• explain the purpose of statistic — the teacher introduces the concept of statistics as a method of collecting and organising information • analyse and interpret statistical data presented in tables — the teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table
Instructional materials	• classroom register, bottle tops, pencils or counters.
Evaluation	• (a) Define statistic • (b) State three methods of collecting data. • (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Lesson objectives

1.	explain the purpose of statistic
2.	analyse and interpret statistical data presented in tables

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Need for statistics: purposes, and the collection and tabulation of data

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: Why We Need Statistics

Think about how much information surrounds you every day. In one morning you might see how many cars pass your house, how many goals your favourite team scored last season, or how much your school spends on books each term. All of these are numbers, but numbers on their own are not very useful. The job of statistics is to turn those scattered numbers into something you can understand and use.

Statistics is the branch of mathematics that deals with collecting, organising, analysing, and interpreting data. Data simply means the information or facts that you collect, usually as numbers. The word statistics can mean two things: the whole process of handling data, or the numbers themselves once they have been organised.

What is the purpose of statistics? The main purpose is to make sense of information so that we can make better decisions. Statistics helps us in several ways:

To summarise large amounts of information. If your teacher lists the marks of all 40 students in your class, that is a lot of numbers. Statistics lets you group them into a table so you can quickly see, for example, how many students scored above 60.
To compare. Statistics lets you compare things fairly, such as the attendance of two classes or the rainfall of two months.
To spot patterns and trends. When data is organised, you can see whether something is increasing, decreasing, or staying the same over time.
To support decisions. Governments, companies, and schools all use statistics to plan. A shop uses sales data to decide which goods to stock; a school uses exam data to decide where learners need more help.
To predict. From organised data, you can often make a reasonable guess about what might happen next.

Collecting and tabulating data. The first step in statistics is gathering the data. You can collect it by counting, by measuring, or by asking questions. Once you have the raw data — that is, the data just as you collected it, in no particular order — you organise it into a table. Tabulating means arranging the data into rows and columns so that it is neat and easy to read. A well-made table is the starting point for all the analysis that comes later.</p>

Worked example 1: Seeing the purpose of statistics.

Suppose your school wants to buy sports kits and needs to know which sport learners prefer. The school asks 20 learners and records each learner's choice:

Football, Tennis, Football, Swimming, Tennis, Football, Swimming, Football, Tennis, Football, Swimming, Football, Tennis, Football, Football, Swimming, Tennis, Football, Swimming, Football

These are the raw data — 20 answers in the order they were collected. On their own they are hard to read. Now let's tabulate them. We count how many learners chose each sport.

Football appears 10 times, Tennis appears 4 times, and Swimming appears 6 times.

<table border="1" cellpadding="5">
<tr><th>Sport</th><th>Number of learners (frequency)</th></tr>
<tr><td>Football</td><td>10</td></tr>
<tr><td>Tennis</td><td>4</td></tr>
<tr><td>Swimming</td><td>6</td></tr>
<tr><td>Total</td><td>20</td></tr>
</table>

The table immediately shows the purpose of statistics: the messy list has been summarised into a clear picture. We can see at a glance that Football is the most popular and Tennis is the least popular. With this organised data, the school can decide how many kits of each sport to buy. Notice also that the frequencies add up to 20, which confirms we counted every learner.</p>

Worked example 2: Reading and interpreting a table.

A farmer records the number of eggs collected from her hens on each day of one week.

<table border="1" cellpadding="5">
<tr><th>Day</th><th>Number of eggs</th></tr>
<tr><td>Monday</td><td>25</td></tr>
<tr><td>Tuesday</td><td>28</td></tr>
<tr><td>Wednesday</td><td>22</td></tr>
<tr><td>Thursday</td><td>30</td></tr>
<tr><td>Friday</td><td>26</td></tr>
<tr><td>Saturday</td><td>24</td></tr>
<tr><td>Sunday</td><td>27</td></tr>
</table>

What can this table tell us? Interpreting means reading meaning out of the organised data.

Thursday had the highest number of eggs, with 30.
Wednesday had the lowest number of eggs, with 22.
The eggs collected each day are between 22 and 30, so the hens are fairly steady producers.

Now suppose the farmer wants to know roughly how many eggs she gets in a week. She can add the numbers: 25 + 28 + 22 + 30 + 26 + 24 + 27 = 182. So in one week the hens produced 182 eggs. This total is itself a useful statistic. The farmer could compare this week's total with next week's to spot whether the hens are producing more or fewer eggs — a trend that helps her plan for selling the eggs.</p>

Practice

1. A class of 15 learners is asked which of three subjects they enjoy most: Mathematics, English, or Basic Science. Their answers, in order, are:

   English, Science, Maths, Maths, English, Science, Maths, English, Maths, Science, Maths, English, Science, Maths, English

   a) Count how many learners chose each subject and write the results in a table with columns for Subject and Number of learners.
   b) Which subject is most popular, and which is least popular?
   c) What is the total number of learners shown by your table? Why must this total be 15?
2. The table below shows the number of books borrowed from a school library each day of one week.

   Day | Books borrowed
   Monday | 40
   Tuesday | 35
   Wednesday | 45
   Thursday | 50
   Friday | 30

   a) Which day had the most books borrowed, and which had the fewest?
   b) How many books were borrowed altogether that week?
   c) Give one way the librarian could use this information to make a decision about the library.
3. In your own words, explain what the purpose of statistics is, and mention at least two ways that organising data into a table helps people understand it.

Section 2: Reading Tables to Make Sense of Data

After data is collected, it is often arranged into a table. A table is a tidy way of showing information in rows and columns so that the numbers are easy to read and compare. Reading a table means looking carefully at its headings to understand what each row and each column stands for, and then pulling out the information you need to answer a question.

Let's look at the parts of a table. The top row (the heading row) tells you what each column contains. The first column usually names the things being counted, and the other columns hold the numbers. To interpret a table, ask yourself three things:

1. What does the table count? (Look at the headings.)
2. Which row or column do I need? (Find the right place.)
3. What does the number actually mean? (Read it with the heading.)

Below is a table showing how 40 students in a school travel to school each morning.

The table below shows how 40 students travel to school.

Mode of travel	Number of students
Walking	18
Bicycle	10
Bus	8
Car	4

We can use this table to answer questions.

Example 1: How many students walk to school?
Find the row "Walking". The number beside it is 18. So 18 students walk.

Example 2: How many more students walk than come by car?
Walking is 18 and car is 4. Difference: $18 - 4 = 14$. So 14 more students walk than come by car.

Example 3: The total of all the numbers is $18 + 10 + 8 + 4 = 40$, which matches the total number of students. This is a quick way to check that the table is correct.

So a table lets us find exact numbers and compare them quickly.

Practice

1. The table below shows the favourite subjects of 30 students in a class.

Subject	Number of students
Mathematics	12
English	6
Science	8
Social Studies	4

   (a) How many students chose Science?
   (b) Which subject did the most students choose, and how many chose it?
   (c) How many more students chose Mathematics than English?

2. A fruit seller recorded the number of fruits he sold in one day.

Fruit	Number sold
Oranges	45
Apples	30
Bananas	25
Mangoes	20

   (a) How many apples did he sell?
   (b) How many more oranges than mangoes did he sell?
   (c) Altogether, how many fruits did he sell that day?

Check your work

Attempt every question before you read this page.

Why We Need Statistics

1. a) Table: Mathematics = 6, English = 5, Basic Science = 4, Total = 15.
   b) Mathematics is most popular (6 learners); Basic Science is least popular (4 learners).
   c) The total is 15, which must equal the number of learners asked, because every learner chose exactly one subject and each was counted once.

   Count each subject from the list. Mathematics appears 6 times, English appears 5 times, and Basic Science appears 4 times. Write these in a table:

   Subject | Number of learners
   Mathematics | 6
   English | 5
   Basic Science | 4
   Total | 15

   Adding the frequencies: $6 + 5 + 4 = 15$. This confirms every one of the 15 learners has been counted exactly once. The largest frequency is 6 (Mathematics), so it is most popular; the smallest is 4 (Basic Science), so it is least popular.
2. a) Most: Thursday (50 books); fewest: Friday (30 books).
   b) $40 + 35 + 45 + 50 + 30 = 200$ books.
   c) Accept any sensible decision, for example the librarian could open earlier or longer on the busiest days, or buy more copies of the kinds of books that are borrowed most.

   a) Compare the numbers in the table: the largest is 50 on Thursday, so Thursday had the most books borrowed. The smallest is 30 on Friday, so Friday had the fewest.
   b) Add all the daily totals: $40 + 35 + 45 + 50 + 30 = 200$. So 200 books were borrowed that week.
   c) The purpose of statistics is to support decisions. The librarian can see which days are busiest and which are quiet, and can use that to plan staffing, opening hours, or which books to stock more of.
3. Accept an answer in the learner's own words. It should state that the purpose of statistics is to make sense of information so that better decisions can be made. It should mention at least two of: it summarises large amounts of data; it makes comparison easy; it helps spot patterns and trends; it supports decisions and predictions. A table helps because it arranges data neatly into rows and columns so it is easy to read and compare.

Reading Tables to Make Sense of Data

1. (a) 8 students chose Science.
   (b) Mathematics; 12 students chose it.
   (c) $12 - 6 = 6$, so 6 more students chose Mathematics than English.

   (a) Read across the "Science" row to the number column: 8.
   (b) Compare the numbers 12, 6, 8 and 4. The largest is 12, which sits in the Mathematics row.
   (c) Subtract the English number from the Mathematics number: $12 - 6 = 6$.
2. (a) 30 apples.
   (b) $45 - 20 = 25$ more oranges than mangoes.
   (c) $45 + 30 + 25 + 20 = 120$ fruits in all.

   (a) Find the "Apples" row and read the number beside it: 30.
   (b) Subtract the mangoes from the oranges: $45 - 20 = 25$.
   (c) Add all four numbers: $45 + 30 + 25 + 20 = 120$.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-63bf0503	**Date**	
Subtopic	Need for statistics: purposes, and the collection and tabulation of data	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____